

# PHYS 5BL–Spring 2026 — Pre-Lab 2 Response

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1. Fiat Lux.

2. (i)  $u = f((x, z)^T)$   
 $\implies \bar{u} \approx f((\bar{x}, \bar{z})^T) + \frac{1}{2} \text{Tr}[H \Sigma_{(x,z)}^T]$   
 $f((\bar{x}, \bar{z})^T) = \bar{x}\bar{z}$ ,  $H = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ ,  $\Sigma_{(x,z)}^T = \begin{bmatrix} \sigma_x^2 & \text{Cov}(x, z) \\ \text{Cov}(x, z) & \sigma_z^2 \end{bmatrix}$   
 $\implies \bar{u} \approx \bar{x}\bar{z} + \text{Cov}(x, z)$   
 $\Sigma_u \approx J \Sigma_{\mathbf{x}} J^T$   
 $J = \begin{bmatrix} \bar{z} & \bar{x} \end{bmatrix}$   
 $\implies \Sigma_u = \bar{z}^2 \sigma_x^2 + 2\bar{x}\bar{z} \text{Cov}(x, z) + \bar{x}^2 \sigma_z^2$
- (ii)  $f(\bar{x}, \bar{z}) = \frac{\bar{x}}{\bar{z}}$ ,  $H = \begin{bmatrix} 0 & -\frac{1}{\bar{z}^2} \\ -\frac{1}{\bar{z}^2} & \frac{2\bar{x}}{\bar{z}^3} \end{bmatrix}$ ,  $\Sigma_{(x,z)}^T = \begin{bmatrix} \sigma_x^2 & \text{Cov}(x, z) \\ \text{Cov}(x, z) & \sigma_z^2 \end{bmatrix}$   
 $\implies \bar{u} = \frac{\bar{x}}{\bar{z}} - \frac{\text{Cov}(x, z)}{\bar{z}^2} + \frac{x \sigma_z^2}{y^3}$   $\Sigma_v \approx J \Sigma_{\mathbf{x}} J^T$   
 $J = \begin{bmatrix} \frac{1}{\bar{z}} & -\frac{\bar{x}}{\bar{z}^2} \end{bmatrix}$   
 $\implies \Sigma_v \approx \frac{\sigma_x^2}{\bar{z}^2} - \frac{2\bar{x} \text{Cov}(x, z)}{\bar{z}^3} + \frac{\bar{x}^2 \sigma_z^2}{\bar{z}^4}$
- (iii) If we let  $y = [xz \ x/z]^T$ , then  $\Sigma_y$  will be:

$$\Sigma_y = \begin{bmatrix} \sigma_u^2 & \text{Cov}(u, v) \\ \text{Cov}(u, v) & \sigma_v^2 \end{bmatrix}$$

$$J = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial z} \end{bmatrix} = \begin{bmatrix} \bar{z} & \bar{x} \\ \frac{1}{\bar{z}} & -\frac{\bar{x}}{\bar{z}^2} \end{bmatrix}, \text{ so:}$$

$$\Sigma_y \approx J \Sigma_{(x,z)}^T J^T$$

$$\approx \begin{bmatrix} \bar{z} & \bar{x} \\ \frac{1}{\bar{z}} & -\frac{\bar{x}}{\bar{z}^2} \end{bmatrix} \begin{bmatrix} \sigma_x^2 & \text{Cov}(x, z) \\ \text{Cov}(x, z) & \sigma_z^2 \end{bmatrix} \begin{bmatrix} \bar{z} & \bar{x} \\ \frac{1}{\bar{z}} & -\frac{\bar{x}}{\bar{z}^2} \end{bmatrix}^T$$

We only care about  $\Sigma_{y21}$ , which is:

$$\begin{aligned}
 \text{Cov}(u, v) &\approx \begin{bmatrix} \bar{z} & \bar{x} \end{bmatrix} \begin{bmatrix} \sigma_x^2 & \text{Cov}(x, z) \\ \text{Cov}(x, z) & \sigma_z^2 \end{bmatrix} \begin{bmatrix} \frac{1}{\bar{z}} \\ -\frac{\bar{x}}{\bar{z}^2} \end{bmatrix} \\
 &= \begin{bmatrix} (\bar{z}\sigma_x^2 + \bar{x}\text{Cov}(x, z)) & (\bar{z}\text{Cov}(x, z) + \bar{x}\sigma_z^2) \end{bmatrix} \begin{bmatrix} \frac{1}{\bar{z}} \\ -\frac{\bar{x}}{\bar{z}^2} \end{bmatrix} \\
 &= (\bar{z}\sigma_x^2 + \bar{x}\text{Cov}(x, z)) \left( \frac{1}{\bar{z}} \right) + (\bar{z}\text{Cov}(x, z) + \bar{x}\sigma_z^2) \left( -\frac{\bar{x}}{\bar{z}^2} \right) \\
 &= \left( \frac{\bar{z}\sigma_x^2}{\bar{z}} + \frac{\bar{x}\text{Cov}(x, z)}{\bar{z}} \right) - \left( \frac{\bar{x}\bar{z}\text{Cov}(x, z)}{\bar{z}^2} + \frac{\bar{x}^2\sigma_z^2}{\bar{z}^2} \right) \\
 &= \sigma_x^2 + \frac{\bar{x}}{\bar{z}}\text{Cov}(x, z) - \frac{\bar{x}}{\bar{z}}\text{Cov}(x, z) - \frac{\bar{x}^2}{\bar{z}^2}\sigma_z^2 \\
 &= \sigma_x^2 - \frac{\bar{x}^2}{\bar{z}^2}\sigma_z^2
 \end{aligned}$$

3.

